The program functionalities:

User:

- Log-in to the system using Username/password.
- You can view all the products.
- Can view the product details.
- And can only submit one feedback for each product.
- View all submitted feedback.
- Can update submitted feedback.
- Can delete his submitted feedback.
- Change your password.

Administrator:

- Log-in to the system using Username/Password.
    - Access to dashboard
    - Dashboards contain charts for live data
    - Total no of feedback submitted for each product
    - Average performance of each product
- Manage Users
    - Check for total no of users
    - Search user
    - Can filter user by Role and status
    - Remove a user
    - Can change role of user
    - Can change status of user
- Manage Products
    - Check for total no of product
    - Search for a product
    - Remove a product
    - Can update an existing product
    - Can add new product
- Manage feedback submitted by users
    - Check total no of feedback submitted
    - Print submitted feedback

    - Remove submitted feedback
- Change your password

BACKEND

- Tools & Technology
    - .NET core 8.0
    - EF core
    - SQL Server
    - Vs code
- Packages
    - dotnet add package Microsoft.EntityFrameworkCore
    - dotnet add package Microsoft.EntityFrameworkCore.sqlserver
    - dotnet add package Microsoft.EntityFrameworkCore.tools
    - dotnet add package Microsoft.EntityFrameworkCore.design
    - dotnet add package FluentValidation --version 11.9.0
    - dotnet add package FluentValidation.AspNetCore
    - dotnet add package System.IdentityModel.Tokens.Jwt
    - dotnet add package Microsoft.AspNetCore.Authentication.JwtBearer
- APIs
- User APIs
    1. Login $
    2. sign-up
    3. Get all users
    4. Update status of users
    5. Update Role of users
    6. Remove user
    7. Change password
- Product APIs
    1. Add new product
    2. Get all products
    3. Get product by ID
    4. Update product
    5. Delete product
    6. Get product by category ID
- Feedback APIs
    1. Submit feedback
    2. Update feedback
    3. Delete feedback
    4. My feedback

5. Get all feedback
6. Get Feedback by ID

- Product Category
  1. Add new category
  2. Update category
  3. Get all category

- Database Tables (code first approach)
  - User table

100 %

Results | Messages

	Id	Name	Email	Password	Gender	PhoneNumber	Role	status
1	1	admin	admin@gmail.com	admin	male	1234567890	admin	true
2	2	mukesh	mukesh@gmail.com	mukesh	male	99876	Customer	false
3	3	string	muk@gmail.com	string	string	123	user	false
4	4	Kumar	kumar@gmail.com	kumar	M	1234	user	true
5	5	ankit	ankit@gmail.com	ankit	Male	1234	user	true

  - Feedback Table

Results | Messages

	Id	Name	Email	ProductId	Rating	Comment
1	2	string	string	1	0	string
2	8	string	kumar@gmail.com	1	10	string
3	9	kumar	kumar@gmail.com	2	5	not a good product
4	10	ankit	kumar@gmail.com	3	7	hi i am ankit

  - Product table

Results | Messages

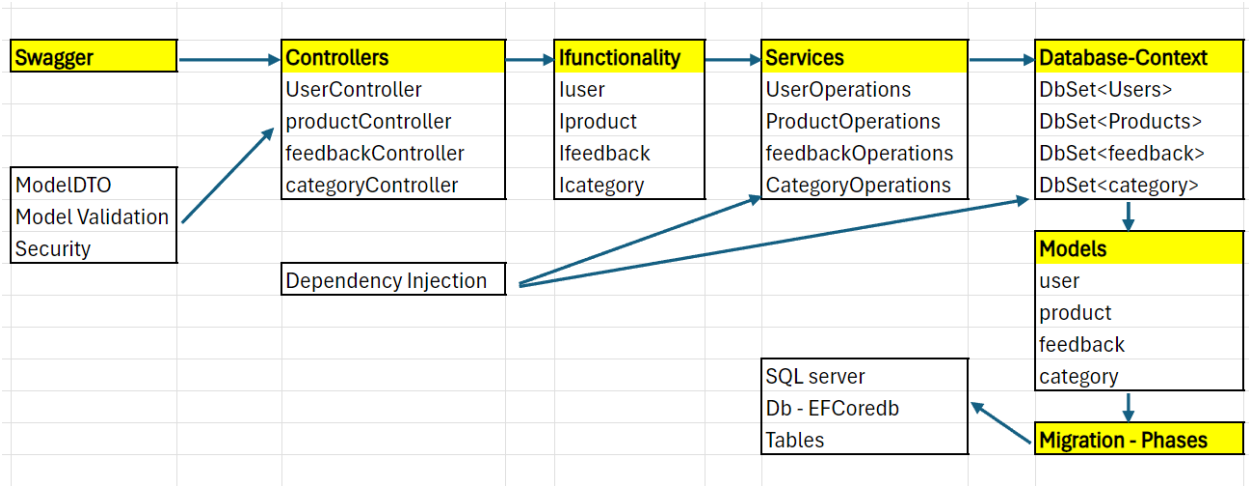
	id	CategoryId	Name	Description	Clients	ImageUrl
1	1	1	XTRAKTO.AI	Unstructured text, image, video, audio data converte...	27	product1.png
2	2	1	EXELIA.AI	Human like interactions through voice & chat, make ...	6	product2.png
3	3	1	PAYMENTOR	AI-powered collections and receivables managemen...	5	product3.png
4	4	1	SMARTAUDIT.AI	Over 10% of cost of operations is consumed by com...	6	product4.png
5	5	1	AUTOMATION	A approach combines the power of Automation, AI ...	38	product5.jpg

  - Category table

Results | Messages

	id	Name
1	1	AI Solutions
2	2	Platform Based Services
3	3	Insurance

- Backend Architecture



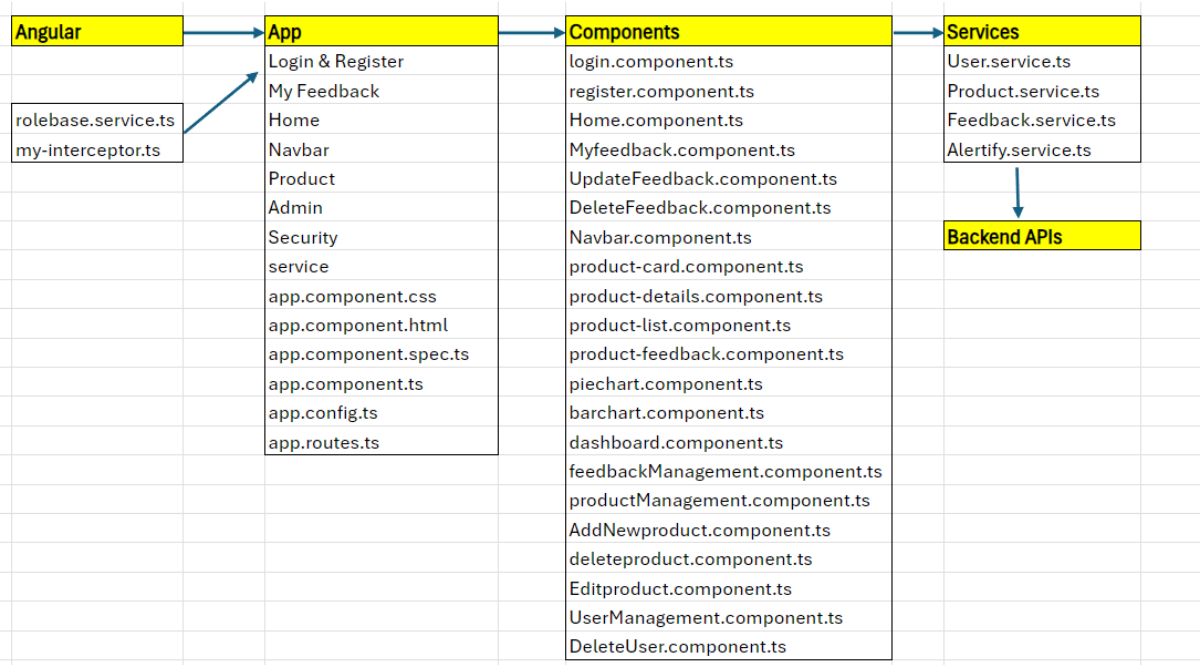
FRONTEND

- Tools & Technology
  - Angular 17.3.0
  - Bootstrap 5.3.3
  - CSS, HTML, TYPE SCRIPT
  - Angular material - responsive, reusable component
  - JWT token base – Authentication and Authorization
  - Rxjs, `alertifyjs,chart.js,jsPDF`

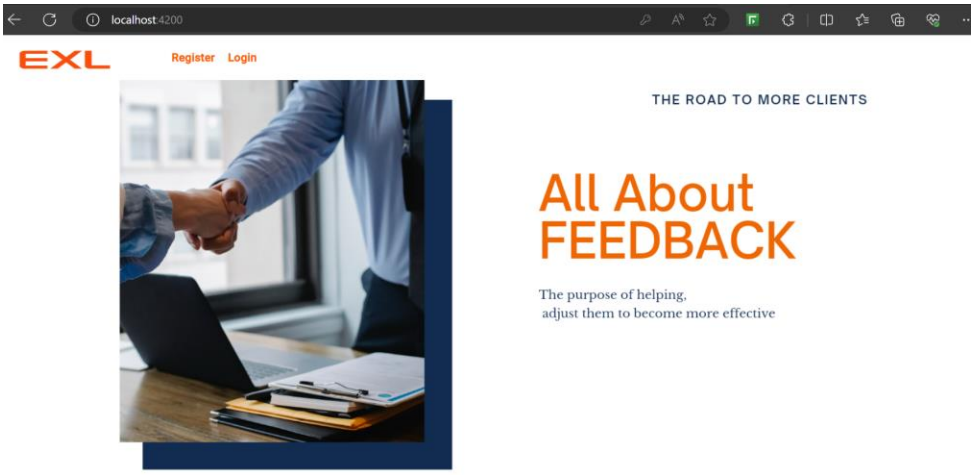
- End points
  - Login user/admin
  - Register
  - Logout
  - Change password
  - Home
  - User Interface
    - My feedback
      - Update submitted feedback
      - Remove submitted feedback
    - Product list
      - Product information
      - Feedback submission

- Admin Interface
  - Dashboard
    - Pie chart
    - Bar chart
  - Manage users
    - Update user role
    - Update user status
    - Remove user
  - Manage products
    - Update product
    - Add new product
    - Remove existing product
  - Manage feedback
    - Remove feedback
    - Print feedback

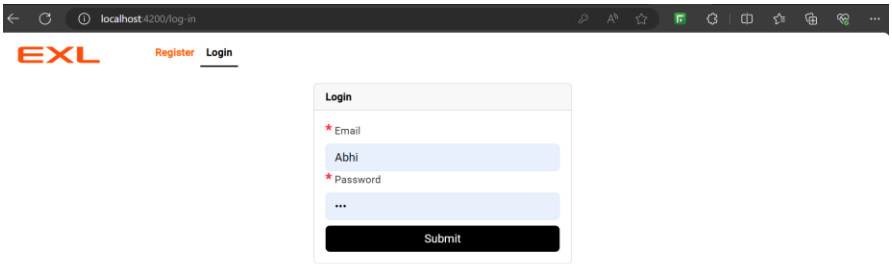
- Frontend Architecture



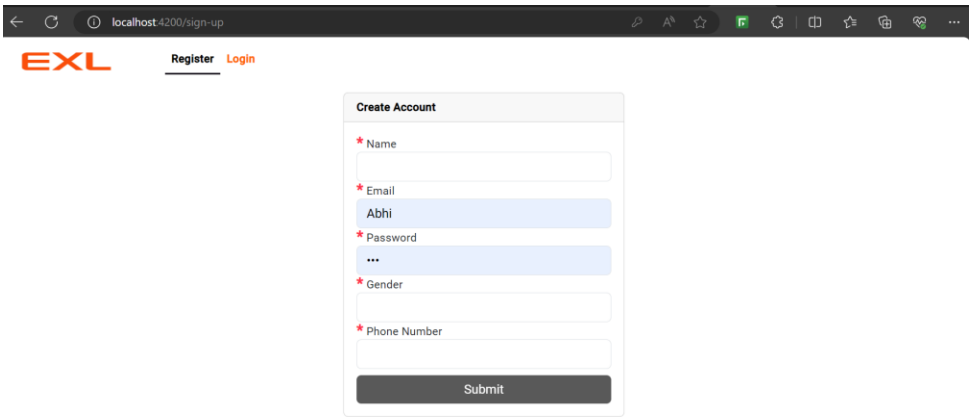
Home Page -



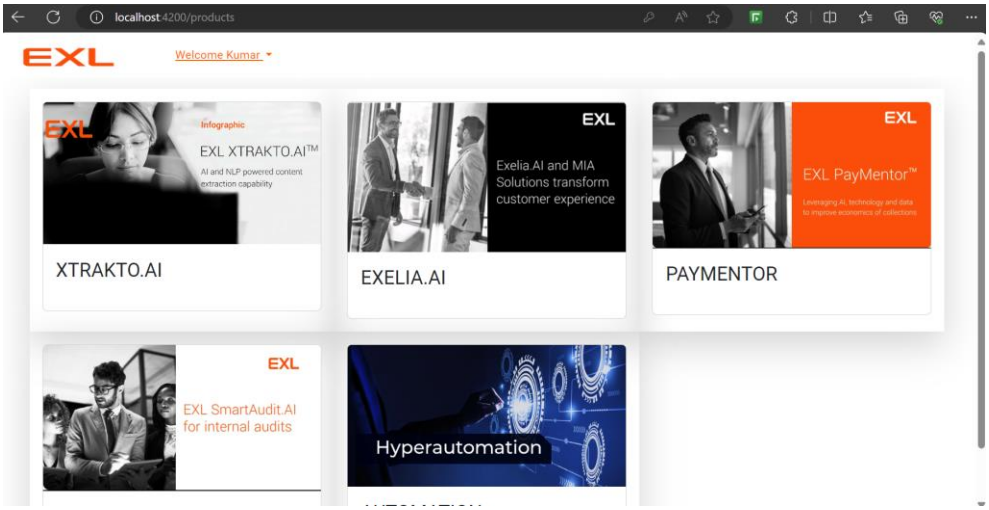
Login -



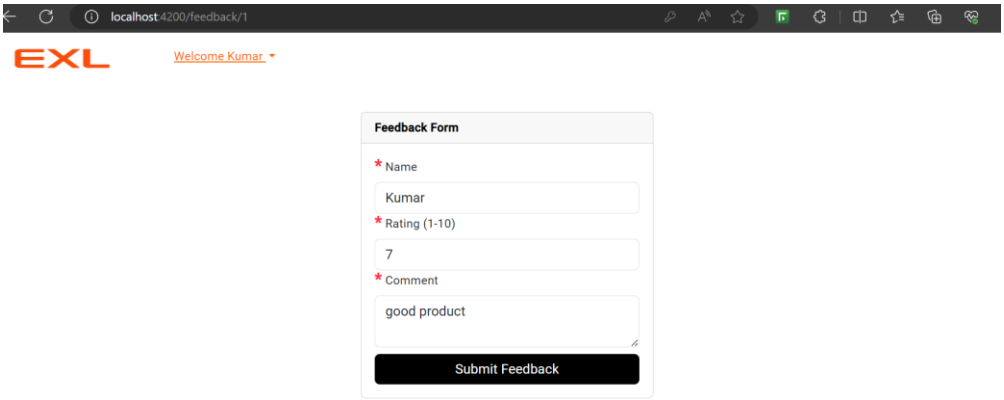
Register -



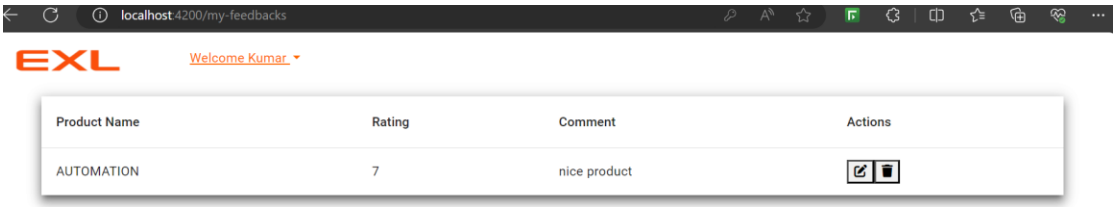
User dashboard -



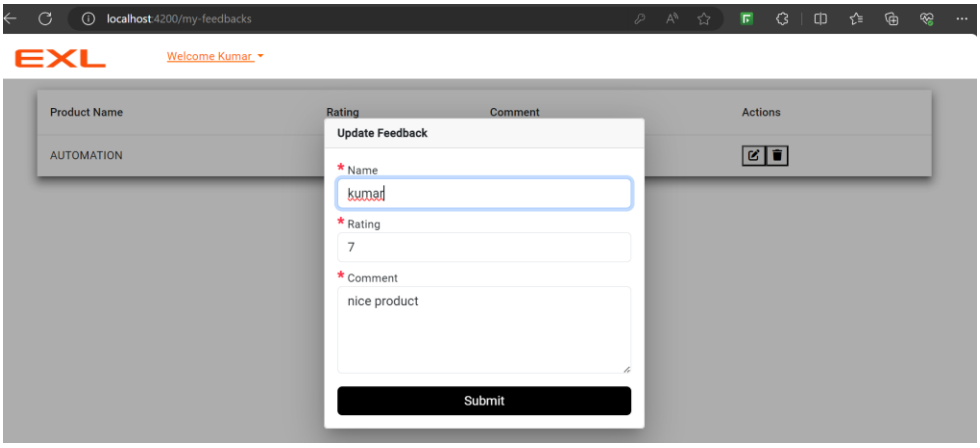
Feedback -



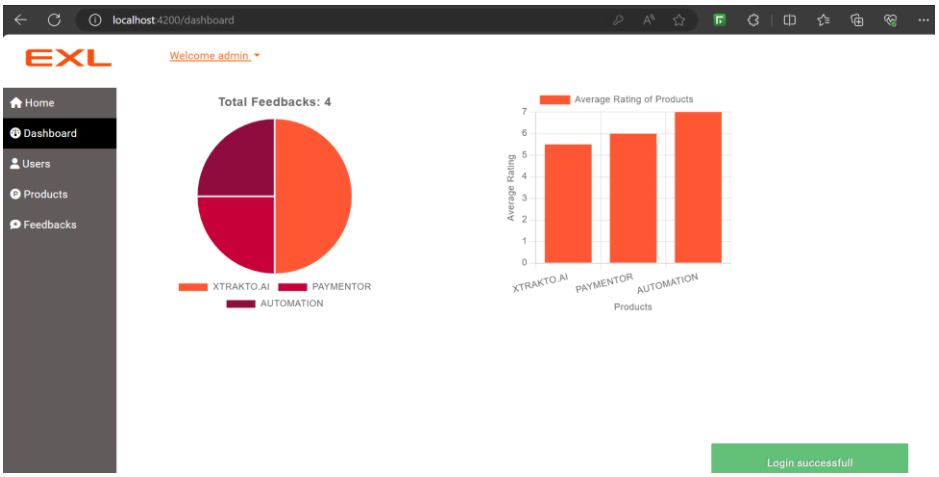
My feedback -



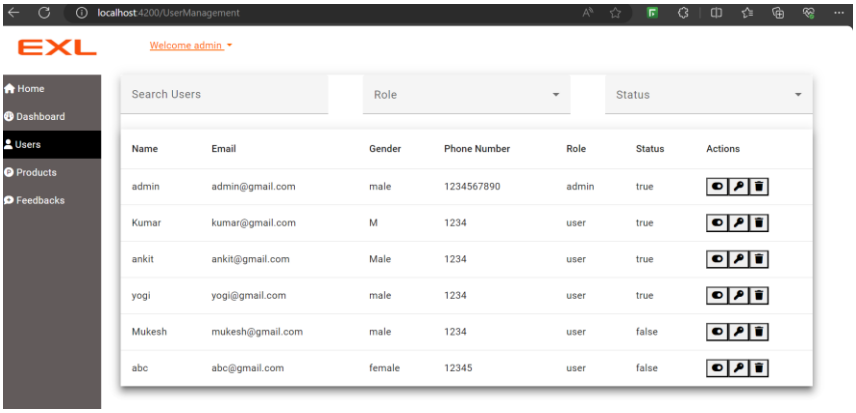
Update Feedback -

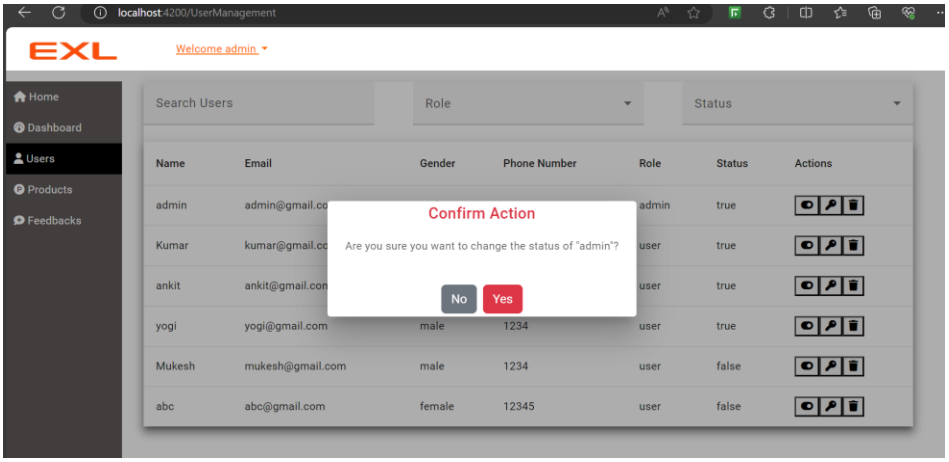


Admin dashboard -

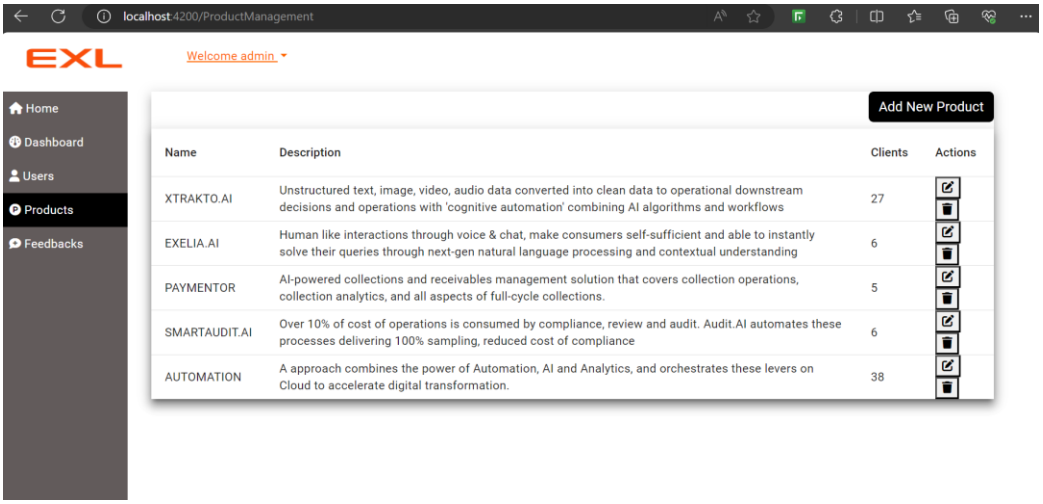


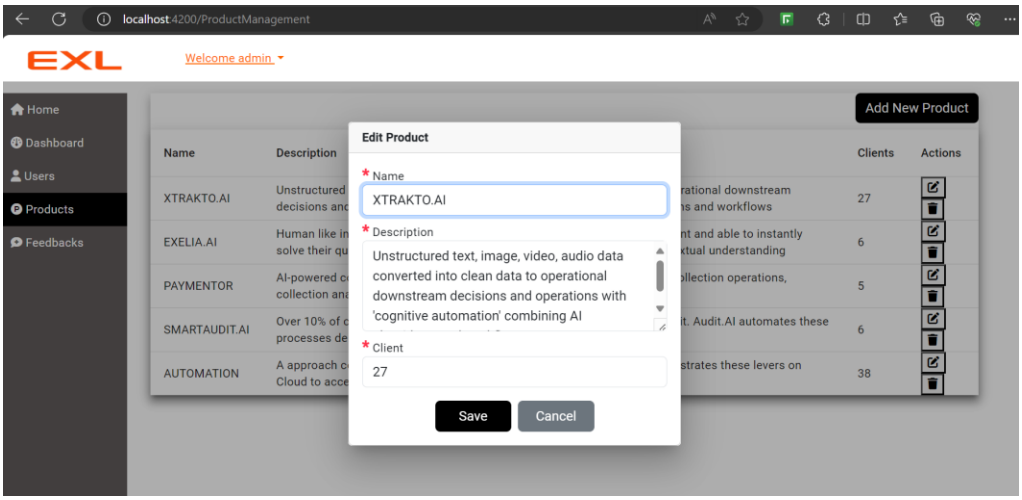
User Management-

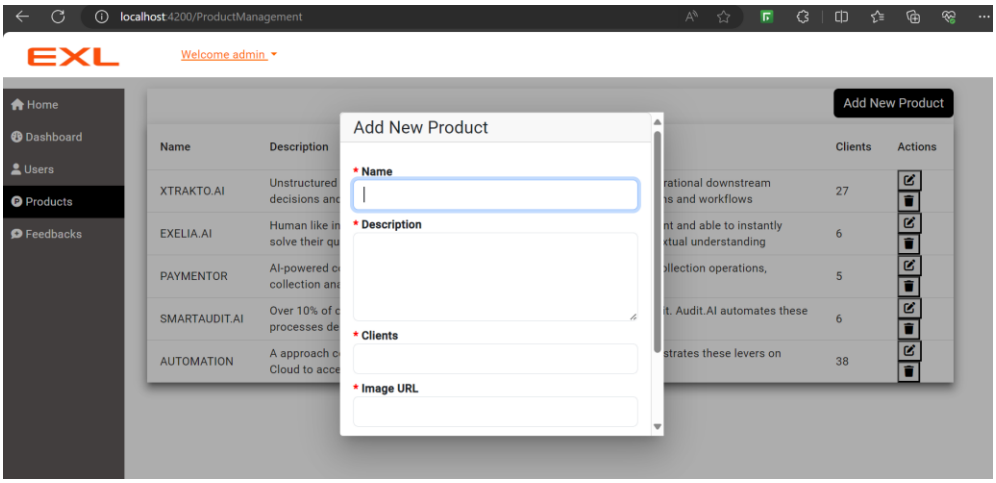


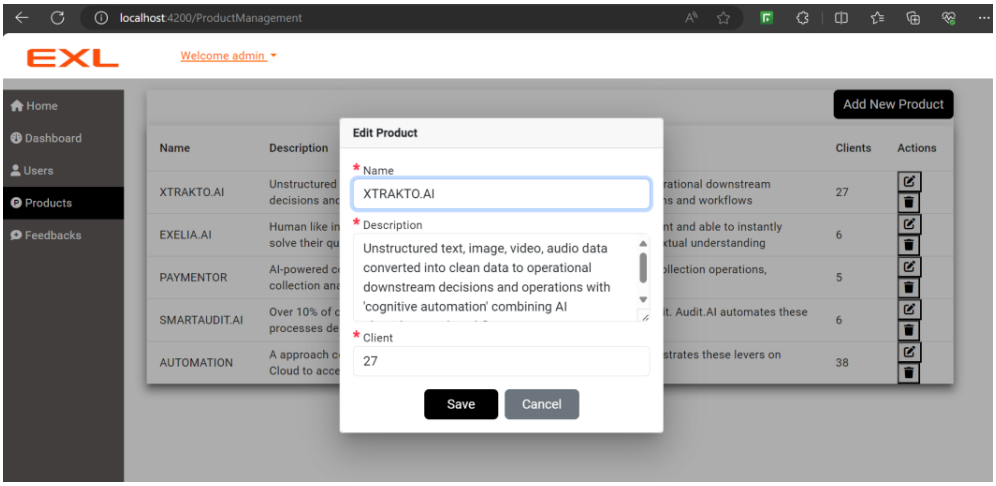


Product Management-

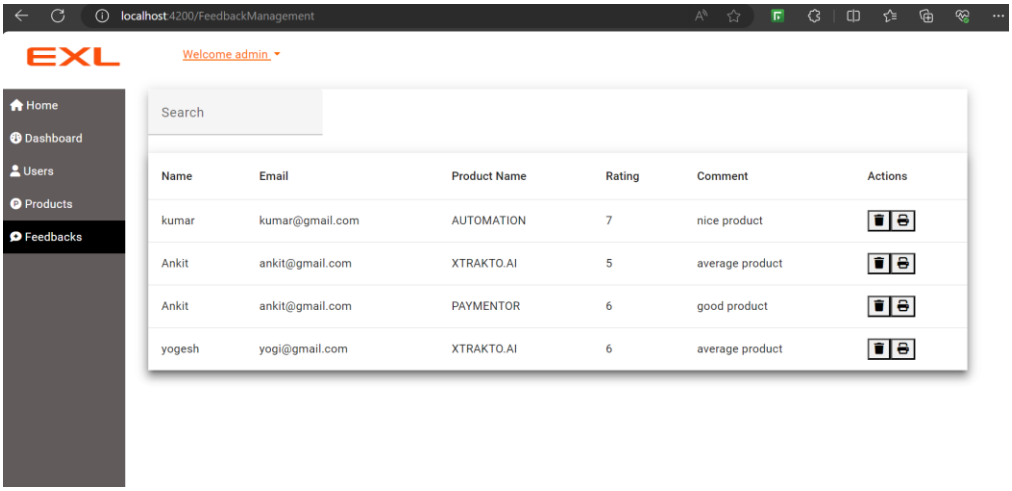




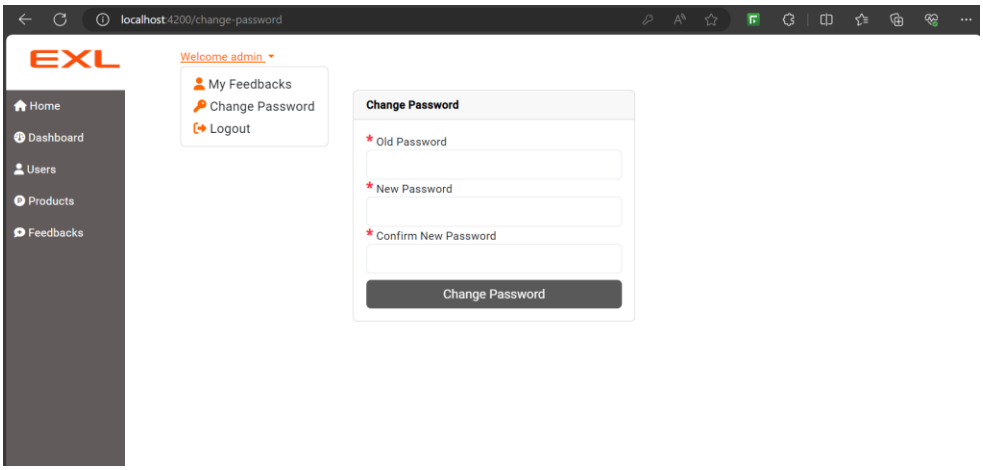




Feedback Management-



Change password -



Top Learnings

- **Backend**
  - Dependency Injection / IOC
  - Model DTO
  - Model Validation
  - Abstraction
- **Security**
  - Authentication and Authorization
  - JWT token management
  - Auth Gard's
  - Interceptor
  - Session management
- **Role Base Access Control**
  - Implementing role-based authorization emphasized the importance of securing sensitive endpoints and ensuring only authorized users can access critical functionalities
- **Frontend**
  - Charts building, pie chart, bar chart using chart.js
  - Printing feedback using jsPDF
  - Used angular material to create responsive data table
  - Alertify services to show alert messages

Future Considerations

Advanced Security Features: Consider exploring additional security measures like two-factor authentication (2FA) and OAuth.

Use SMTP to implement forgot password.

Performance Optimization: As application scales, think about optimizing database queries and improving performance

User Experience: Enhancing user interfaces to provide clear feedback on access restrictions and permissions.